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Review

The Default Mode Network and inner time consciousness

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The default mode network (DMN) exhibits distinct dynamic and topographic features relative to the brain's other networks, yet their link to consciousness remains unclear. We review both neural (DMN-based topography and dynamics) and mental (experience of time speed, mental time travel, and self-non-self) findings across diverse nonordinary states of consciousness, including reduced (anesthesia, Unresponsive Wakefulness Syndrome), elevated (meditation and, to some degree, psychedelics), and abnormal (depression, mania) consciousness. Both reduced and elevated states are featured by topographic 'flattening' of the brain's DMN-centric organization while they differ in their dynamics, that is, longer versus shorter timescales. A neural continuum emerges with different degrees of DMN-centric topography along its extremes: from heightened DMN-centric re-organization in depression, over flattened configurations in psychedelics and meditation, to heightened sensorimotor-centric re-organization in mania. This neural continuum parallels a mental continuum along different balances within inner time consciousness, including slow versus fast time speed, past–present–future (mental time travel) and self versus non-self/other. In conclusion, we propose novel neurophenomenological hypotheses about the intrinsic relationship ('complex correspondence') of the brain's DMN-centred topography and dynamics with inner time consciousness.

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Introduction

The default mode network (DMN) possesses peculiar spatial (topographic) and temporal (dynamic) features, which distinguish it from virtually every other brain network [8,33,45]. Spatially, the DMN exhibits remarkable functional (and structural) connectivity characteristics: it demonstrates high within-network connectivity and occupies a position at the apex of the cortical hierarchical gradient, which puts this network at the opposite end of sensorimotor regions/networks [3,29,33]. Temporally, the activity of the DMN exhibits peculiar dynamic features: it shows stronger power in the infraslow frequency range (0.01–0.1 Hz; as measured with fMRI) [26,52]; moreover, it also shows higher power in faster frequencies, as in the gamma frequency band [43,34], higher neural variability [30], and longer intrinsic neural timescales (INTs). INT durations show high variability across brain networks; they are measured through the computation of the signal's temporal autocorrelation function, as operationalized by the autocorrelation window) in both infraslow (0.01–0.1 Hz functional magnetic resonance imaging [fMRI] [23]) and faster (1–80 Hz; electroencephalography/magnetoencephalography; [18,48]) frequency ranges.

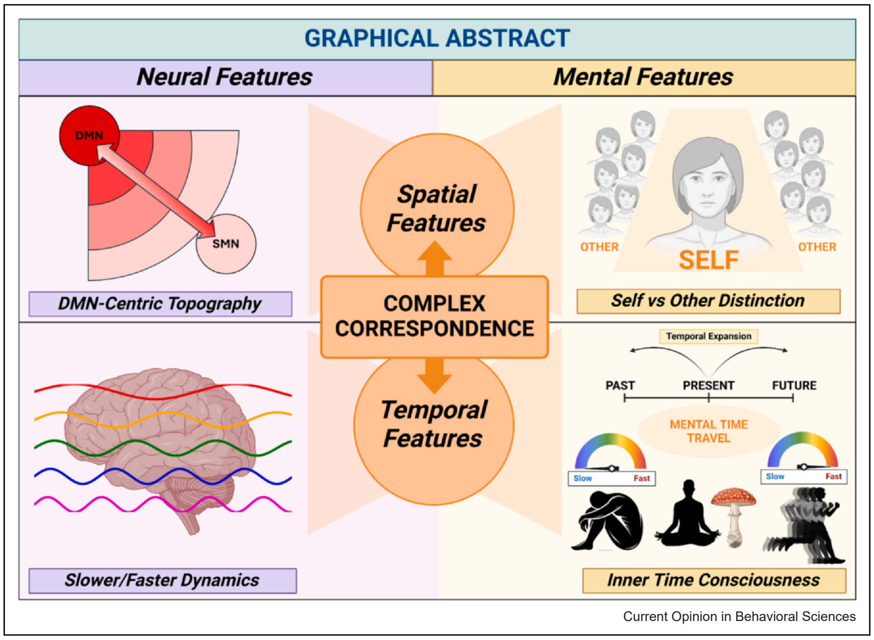
Are these spatial and temporal features of the DMN related to consciousness? The goal of our paper consists in demonstrating that the DMN's specific spatial-topographic and temporal-dynamic features stand in an intrinsic relationship (a 'complex correspondence', [36]) to one key feature of consciousness, namely, inner time consciousness which by itself can be characterized by the experience of slow-fast speed, past-present and future, and self-non-self balances (see below for details; Figure 1). Is the DMN related to inner time consciousness and its three key phenomenal features? Addressing this question, we review recent findings on the topography and dynamics of the DMN in a variety of different states of consciousness — this serves the goal to develop specific neurophenomenological hypotheses. That is complemented by considering the original role of the DMN as baseline or default of the brain (Raichle; Box 1) and the calculation of its topographic-dynamic gradients (Box 2).

Default mode network and global states of consciousness

Spatial features — loss of default mode network–centred topographic organization in both reduced and elevated states of consciousness

A recent review on psychedelics [50], which includes data from studies on ayahuasca, psilocybin, lysergic acid

Figure 1



Graphical abstract. The connection of neural and mental features through their underlying spatial and temporal dimensions. On the neural side, the DMN and SMN reveal a DMN-centric topography with a gradient from DMN-based slower to SMN-related faster neural dynamics. We can see analogous or corresponding features in consciousness, specifically inner time consciousness. Following the findings, inner time consciousness can be characterized by the experience of (i) slow-fast time speed, (ii) past-present and future (mental time travel), and (iii) self-non-self distinction. The complex correspondence of neural and mental activity in their underlying spatial and temporal features bridges and connects them and are thus shared by both as their 'common currency' ([41], 2024).

Box 1 Returning to Raichle — Does the DMN serve as internal neuronal baseline or default for both brain and consciousness?

Despite a recent rise in interest in many topographic and dynamic aspects within different leading theories of consciousness, the DMN's role for consciousness has not seen a parallel increase in popularity. The Integrated Information Theory of consciousness highlights the posterior cortical regions as key for consciousness [2], while the Global Neuronal Workspace theory focuses on the prefrontal cortex, namely, the dorsolateral prefrontal cortex [32]. Others like the higher-order thought theory and the attention schema theory highlight the prefrontal cortex and the attention networks [6]. This leaves open the role of DMN for consciousness which we here address in our paper.

This changes once one considers a view of both brain and consciousness that takes into account the spatial and temporal characteristic of neural activity of the brain as a whole, such as in the temporo-spatial theory of consciousness (TTC) [42]. TTC, among its proposed mechanisms for consciousness, presupposes the whole brain's neural activity and its global topography within which the DMN plays a key role (since it stands at the apex of the brain's hierarchical organization) — this predisposes it to serve as baseline, that is, as internal standard or reference for the rest of the brain's neural activity [40]. Such role of the DMN, serving as internal neuronal baseline for the rest of the brain, can be traced to Marcus Raichle himself and his original conception of the DMN as default [45].

What are indices of the DMN serving as internal neuronal baseline as default for the rest of the brain? Among others, these include (i) anticorrelation of the DMN with other networks, (ii) the DMN's task-unspecific deactivation, (iii) the DMN's stronger causal output to other networks than in the reverse direction, and (iv) modulation of faster activity in other networks by the DMN through its own slower activity [33]. Despite their differences, these four features share that they index the DMN's role as internal neuronal baseline or default for the rest of the brain. Importantly, these indices seem to be lost during the loss of consciousness [13] or abnormally altered in psychedelic and meditative states — this suggests that the DMN may be key for consciousness and its specific configurations (like its inner time consciousness with its three features) through the degree to which it takes on the role as internal neuronal baseline or default for the rest of the brain.

In conclusion, we propose that the DMN, through its specific topographic and dynamic features relative to the rest of the brain, may serve as internal neuronal baseline, default, reference, or standard for the brain's cortex as whole and its associated cognitive functions. Given the intimate connection of changes in DMN topography and dynamics with changes in both the global state of consciousness and inner time consciousness (see main text), we tentatively propose that the DMN's role as neuronal baseline or default may be one potential mechanism through which it is linked to consciousness.

Box 2 How are cortical gradients computed? And what do we mean by 'flattening'?

Cortical gradient analysis is a computational approach that reduces complex, high-dimensional brain connectivity data into one or more continuous axes. At the whole-brain level, gradients delineate the global interrelations among distributed functional systems and thus have the power to illustrate how the spatial characteristics of individual systems are embedded within broader organizational patterns. Cortical gradients are computed by reducing high-dimensional brain data — such as measures of FC, structural connectivity, or microstructural properties — into one or more continuous axes that capture gradual spatial transitions across the cortex. Commonly, methods such as PCA, diffusion embedding, Laplacian eigenmaps, or other dimensionality reduction techniques are applied to connectivity matrices to extract these gradients, revealing systematic variations from unimodal sensory regions to transmodal association areas. For example, as shown by Ref. [29], a principal gradient extends from unimodal sensory and motor regions to transmodal, integrative areas — such as the DMN. Various studies show that the DMN stands at the top of the gradient hierarchy, that is, at its *apex*, while sensorimotor regions can be found at its lower end. The DMN thus takes on a special role in the brain's gradient-based topographic hierarchy in terms of its high FC patterns, longer neural timescales, and excitation–inhibition balance titled toward high levels of excitation. It is important to underline that the cortical topography of neural features has also been shown to fluctuate (see Ref. [7] for a recent example). How about the flattening of the unitransmodal gradient we postulate in this work? A significant reduction in the correlation between a gradient obtained from actual functional data and an established cortical hierarchy map may help indicating a phenomenon that here we describe as a 'flattening' of the hierarchy, which is meant as a reduction in the differentiation between unimodal and transmodal regions suggesting that the default mode network no longer occupies its typical apex position. Similarly, joint embedding procedures can help identify deviations from ordinary cortical gradient topographies, detecting interindividual variation. Therefore, as we propose in our paper, since deviations in cortical gradient topography arise in nonordinary states of consciousness, such as meditation and psychedelics, we propose the degree of nonhierarchy of uni- and trans-modal regions relates to a loss of the distinctions of past-future, self-non-self, and slow-fast speed in inner time consciousness.

diethylamide, and 3,4-methylenedioxymethamphetamine, draws an interesting link between DMN functional connectivity (FC) and subjective experience. They show a consistent reduction in within-DMN FC across different psychedelics, which is linked to several mental features like decreased mental time travel into the past, ego dissolution, and reports of increased intensity of subjective experience [9,50]. Analogous fMRI-based reductions in within-DMN FC and changes in time experience can also be observed in other altered states of consciousness as, for instance, during proficient meditation across different traditions [12]. However, similar reduced DMN activity and within-DMN FC have also been observed in fMRI at the opposite end of global states of consciousness, that is, in decreased global levels of consciousness, as for instance in Unresponsive Wakefulness Syndrome (UWS) [20,21] and during absence seizures [19]. The aim of this paragraph is to propose a solution to these apparently contradictory findings.

As already stated, the DMN has a singular manner of relating to other brain regions. In fact, one of its trademark features is its 'anticorrelation' with other networks' activity, such as the central executive network (CEN), the dorsal attention network, and the sensorimotor network (SMN), which is believed to reflect a functional distinction between internally and externally directed thought [13]. Interestingly, these DMN-based anticorrelations, observed in fMRI, are reduced in a wide range of reduced states of consciousness (see Ref. [13] for a thorough review on the subject); on the other hand, these DMN anticorrelations become undetectable during severe loss of consciousness in deep anesthesia [22] and in unresponsive brain-damaged patients [14].

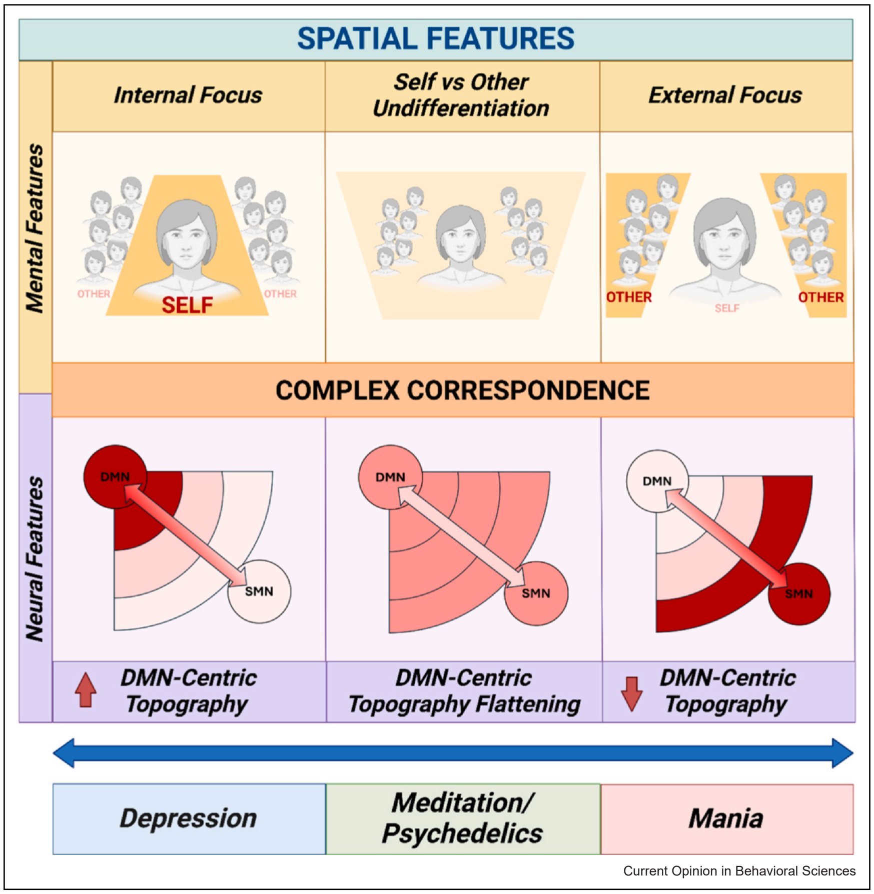
However, psychedelics may not induce elevated states of consciousness in all of its features [4]. An even better example for an elevated state of consciousness is meditation, where the aforementioned loss of the characteristic DMN anticorrelations (in fMRI) can also be observed [12,50]. For the sake of simplicity and in accordance with the somewhat similar findings in their DMN topography as in meditation (see also [47]), we will subsume psychedelics under the umbrella of the elevated states of consciousness.

Together, these findings show that the DMN 're-organizes and de-differentiates' its topographic role within the brain's global spatial organization in altered states of consciousness. Re-organization refers here to the role of the DMN in the global brain's topography, that is, how it stands in relation to other neural networks like the CEN, the salience network, and SMNs (and others). While dedifferentiation points to a particular form of re-organization: the DMN is no longer distinguished (i.e. differentiated) from the other neural networks in both its spatial (in the extent of its within-DMN and between-network FC) and temporal (in the duration of its INTs) features — the DMN does not 'stand out' anymore in both reduced and elevated states of consciousness. This is different in the 'normal' awake conscious states where the DMN is situated at the apex of the brain's topographic-dynamic hierarchy ([18,25,29] and b). Together, these findings suggest a key role for the degree of the brain's DMN-centric topographic-dynamic re-organization for the sustenance of consciousness, in both its reduced and elevated states (Figure 2).

Temporal features — default mode network slow and fast dynamics distinguish between reduced and elevated states of consciousness

The DMN shows slower dynamics with longer dwell durations in fMRI [15] and a concomitant decreased dynamic repertoire of brain states during both anesthesia and UWS, which goes along with a disruption of the

Figure 2



Spatial features of neural (DMN topography) and mental (self vs non-self) states showing complex correspondence across different states of consciousness, including bipolar disorder. Both reduced (disorders of consciousness like coma) and elevated (meditation, psychedelics) states of consciousness are characterized by the flattening of their brain's DMN-centric topography. In contrast, bipolar disorder shows higher or extreme degrees of DMN-centric topography corresponding on the mental side to high degree of internal self-focus typical of depressive states, while a topography shifted toward the SMN corresponds to an external focus toward the environment during manic episodes.

hierarchical role of the DMN [26,51]. Furthermore, in fMRI, Klar et al. [26] showed that the loss of this DMN-centered topography organization in anesthesia goes along with a loss of the regular scale-free dynamics, indexed by the Power Law Exponent of the DMN signal during both rest and task states (see also Ref. [51]).

In another fMRI study, Ref. [25] investigated the changes in INT lengths during rest and task states in sleep: they observed longer INTs in the DMN during N2 sleep in rest, which then significantly shortened during the task state — even during sleep. Furthermore, computational modeling showed that the prolonged INT lengths at rest relate to slower neuronal speed, which can be modulated and shorted by external stimulation ([25] and b), This suggests a putative mechanism for these empirical findings: more prominent slow dynamics — at rest — in N2 sleep in DMN, which, though, can still be modulated and thus be fastened (resulting in shorter INTs) by external stimuli during task states. In this sense, N2 sleep would differ from the anesthetic state, where no evidence for such rest–task modulation could be found [26,51].

The converse occurs in psychedelic states. Findings in mainly electroencephalography/magnetoencephalography show that the speed, variability, and entropy of the brain's neural activity increases within the DMN, which exhibits faster dynamics and exerts less top-down modulation to other networks lower in the hierarchy in such states [50]. This converges with the REBUS model: faster frequencies mean that the speed of the brain's neural activity is faster, which, in turn, implies a higher and faster rate of change. Applied to the cognitive level, this means that thoughts and beliefs undergo more rapid and thus faster changes entailing that no specific beliefs or thought contents stay over longer time and can exert strong top-down modulation of sensory input processing. While, at the same time, the usually slower and more powerful top-down influence, the ongoing thoughts and beliefs, are reduced if not abolished (as in the nondual meditative state; [12]). Together, following the REBUS Model [10], this leads to more relaxed beliefs, less internal constraints with stronger and faster bottom-up influence of the sensory input on information processing during the psychedelic state.

Together, these findings demonstrate a key role of DMN dynamics for the sustenance of global states of consciousness along a continuum from reduced over balanced to elevated states. Total or marginal absence of consciousness is featured by reduced within-network flexibility, with longer dwell durations and reduced network switching; this is complemented by loss of within-DMN dynamics (i.e. loss of scale-free dynamics), prolonged INTs, and decreased neural variability. Altogether, this indicates abnormally slow DMN dynamics when consciousness is reduced or lost (Figure 3).

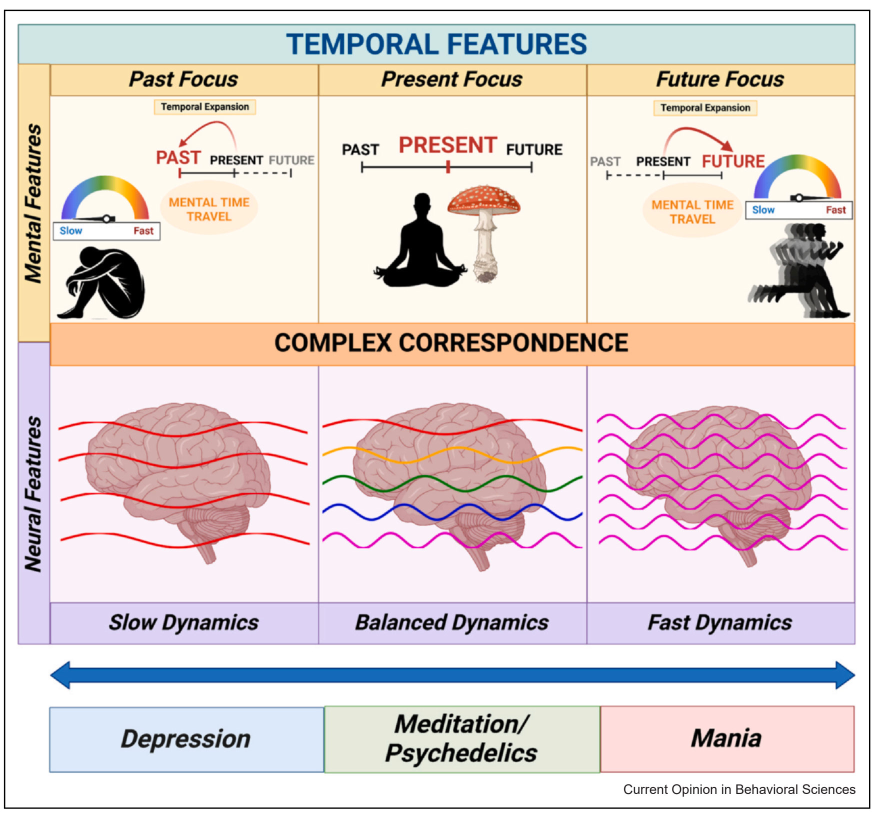
In contrast, in elevated or altered consciousness (as in meditation or psychedelic states), DMN is characterized by faster dynamics with better slow-fast balance: this is manifest in higher within-network flexibility, preserved scale-free dynamics with a shift toward faster frequencies, shorter INT, and higher neural variability (Figure 3). Accordingly, unlike its spatial-topographic characterization, the temporal and thus dynamic features of the DMN allow distinguishing reduced (sleep, anesthesia, coma) and elevated (meditation, psychedelics) global states of consciousness.

Default mode network and Inner time consciousness

From default mode network–centred topography–dynamics to inner time consciousness

The findings show a key role of DMN topography and dynamics in maintaining consciousness. At a finer grain, which phenomenal features of consciousness are shaped by the DMN? The DMN is involved in different forms of internally oriented cognition, including mental self [44], mental time travel [1,46], language-semantics [33], mind wandering [11], and social cognition [8]. Given that, as we said in the previous part, the temporal

Figure 3



Temporal features of neural (DMN dynamics) and mental (inner time consciousness) states showing complex correspondence across different states of consciousness including bipolar disorder. Nonordinary states of consciousness, such as meditation or psychedelics, are characterized by balanced neural dynamics and a focus on the present moment with no (or minimal [virtual and mental]) expansion into past and future (mental time travel). In contrast, in bipolar disorder, slow neural dynamics align with the past-focus and slower inner time speed typical of depressive episodes, while fast neural dynamics correspond to the future-oriented focus and faster inner time speed observed during manic states.

features of DMN activity allow for better distinction between different states of consciousness, we focus our phenomenal analysis specifically on temporal phenomenal features — i.e. the DMN's role in inner time consciousness and specifically mental time travel.

Mental time travel refers to our ability to extend our awareness beyond the present moment, mentally revisiting the past through memory and envisioning the future through anticipation [46]. This inner time consciousness, the subjective experience of time within our minds, stands apart from the outer time of the environment and our perception of it [41]. In mental time travel, the cortical midline regions of the DMN are strongly recruited (as shown in fMRI) during both past and future [46]; whether they are, despite such shared neuro-anatomical basis in the DMN midline regions, mediated by distinct underlying neuronal mechanisms like 'reactivation of memories' (past) and 'prospective memories' (future) remains yet unclear.

Notably, and important in the present context, the same cortical midline regions of the DMN are also involved in the mental self and its self-reflection ([44,35]); this suggests a close relation of our sense of self with inner time consciousness. Specifically, it lends support (albeit indirectly) to a connection of the subjective nature of the self with the temporal continuity of our experience across past, present, and future — the DMN-based self may be inherently temporal and continuous: "I further posit that the DMN integrates component processes and broadcasts them globally to construct and maintain an experience of subjective continuity" [33].

Together, these (and other) observations suggest that inner time consciousness is, among others, characterized by three key phenomenal features: (i) experience of slow versus fast time speed, (ii) experience of past versus future versus present, and (iii) experience self versus non-self. The distinctions within each of these three features are not mutually exclusive but occur in a balanced way thus coming with degrees in our experience of a 'stream of consciousness'. The empirical findings suggest a close relationship between these features of inner time consciousness and the topographic and dynamic features of the DMN. How are these two related?

Flattening of the default mode network–centric topography and dynamics — balancing inner time speed, mental time travel, and self-non-self

Ref. [27] used ultrasound stimulation to causally interfere with activity in the posterior cingulate cortex (PCC), leading to decreased FC among the midline regions of the DMN, (i.e. medial prefrontal cortex and PCC); this was associated with subjects' reports of changes in their time experience ('I saw events from the past', 'My sense of time was distorted'). Furthermore, subjects described higher degrees of mindfulness and decreased global vigor in their consciousness, which, similarly, can also be observed in both meditative [12] and psychedelic states [10].

Many reports indicate a close relationship between changes in DMN activity with subjective experience during altered states of consciousness caused by psychedelics [50]. Decreases of within-DMN FC — mainly within midline DMN regions-, and increased FC from DMN to other brain networks are associated with the reduced ability to perform mental time travel with a stronger focus on the present (rather than past or future). These temporal effects allegedly related to changes in DMN activity are accompanied by changes in the sense of self, namely, ego-dissolution, with decreased duality of self and non-self [12], suggesting again a close relationship of self and time experience [38,49]. Finally, the psychedelic effects on time and self were relatively specific for the DMN, as they could not be observed in relation to psychedelic-related changes in other networks, like in the salience network and the thalamus [50].

The close association of reductions in both within-DMN activity/FC and between DMN-network FC (usually measured in fMRI) with decreased mental time travel and decreased (virtual and mental) expansion of time beyond the present moment are further supported by meditation. During many meditation practices, a strong decrease in DMN neural activity can be observed, along with decreased posterior DMN intraconnectivity, decreased DMN-CEN anticorrelation [16,31] and faster dynamics during both rest and task states [28]. Together with a diminished or absent sense of self (i.e. nondual experience; [12,24]), these changes go along with an altered experience of time, namely, decreased mental time travel into past and future, the experience of a temporal standstill with no flow of time, and an extended present moment experience [5].

Opposite extremes in default mode network–centric topography and dynamics — opposite extremes in inner time speed, past-future mental time travel, and self-non-self distinction

Both psychedelic and meditative states show that a non-DMN-centric topography and faster dynamics modulate inner time consciousness, that is, its three balances of slow-fast speed, past-future, and self-non-self. The flattening of the DMN-centric hierarchical topography and dynamics in these states must be distinguished from its extreme imbalance toward either the DMN or the opposite direction toward the sensorimotor regions/networks in other states as in bipolar disorder with depression and mania — how does that affect these subjects' inner time consciousness?

Depressed subjects with bipolar disorder show a strong shift (in fMRI) toward an even more DMN-centric topography and slower DMN dynamics [30,37]. This is accompanied by experience of an abnormal slowness of inner time, together with predominance of the past, at the expense of the ability to predict future events, and an increased focus on the own mental self. The reverse can be observed in manic BD states: the topography is less DMN centred, shifting away from the DMN toward the opposite extreme, namely, sensorimotor [30,39]. These subjects experience an abnormally fast inner time with focus on the future (at the expense of the past) and an increased focus on and attachment to the outer environment (which co-occurs with a decreased focus on the own inner mental self) (Figures 2 and 3) [17,39]).

Together, the findings show that opposite imbalances of the DMN-centric topography and dynamics toward either the DMN (bipolar depression) or sensorimotor regions (bipolar mania) lead to opposite changes in inner time consciousness, that is, slow versus fast inner time, past- versus future-focus, and self- versus environment-focus. These imbalances of the DMN-centric topography and dynamics toward either of its extreme ends must be distinguished from its flattening with the reduction or elimination of any topographic-dynamic centrism toward either the DMN or SMNs altogether; this can be observed in psychedelics and meditation. Why the shifts toward either slow or fast DMN neuronal speed are associated with opposite changes in their mental time travel, toward either past or future, remains yet unclear: while this is a solid neuro-phenomenological observation, its underlying neuronal and computational mechanisms still remain a matter of future investigations.

Psychedelic and meditative states show decreased (and ultimately abolished) topographic–dynamic distinction of DMN and sensorimotor regions on the neural level. Correspondingly, the phenomenal distinctions of slow versus fast inner time, past versus future, and self versus environment are also reduced if not abolished altogether (as in proficient meditators and extreme psychedelics states); these are then replaced by the experience of 'standstill of time', 'present moment' or 'specious present', and nonduality (of self and environment). Overall, these findings support the assumption that the different degrees of DMN-centric topography and dynamics along a continuum from DMN-centric over noncentric to sensorimotor centric are related and correspond (in a 'complex' rather than 'simple' way; [36]) to a mental continuum of three key dual distinctions (slow vs fast, past vs future, self vs non-self) featuring inner time consciousness (Figures 2 and 3).

Conclusion — complex correspondence of default mode network–centred topography–dynamics and inner time consciousness

Reviewing recent findings, we propose a novel neuroscientific framework about the influence of the DMN on consciousness, explicitly linking neural and mental features.

First, a flattened noncentric DMN spatial and temporal topography is observed in both reduced (coma, anesthesia, sleep) and elevated (psychedelics, meditation) states while their dynamics (slow vs fast) distinguish both: the slow-fast dynamics is balanced in the elevated states while it is abnormally shifted toward the slow dynamics in the reduced states.

Second, remaining within the realm of consciousness, we can observe a neural continuum of different forms of DMN topographic–dynamic re-organization from extreme DMN-centric (bipolar depression) over flattened/noncentric (psychedelics, meditation) to sensorimotor centric (bipolar mania).

Third, this neural continuum of the different forms of the DMN's topographic-dynamic re-organization relates to a corresponding mental continuum along the balances of three key features of inner time consciousness, namely, slow versus fast inner time speed, past versus future mental time travel, and self versus non-self distinction.

Together, our novel neuro-phenomenological framework shows how different forms of the DMN's global topographic and dynamic re-organization lead to corresponding mental re-organization of inner time consciousness in three of its key features, slow versus fast, past versus future, and self versus non-self. We therefore propose 'complex correspondence' of neural and mental features (as distinguished from their 'simple correspondence'), that is, of the DMN's topography–dynamic neuronal organization with the mental organization of inner time consciousness [36]. A definitive explanation of the neuromental mechanisms connecting the DMN to consciousness remains yet an open matter, though. Nonetheless, we here propose a novel neuro-phenomenological framework, which, alongside recent advances in the field (summarized in Box 1), could provide valuable insights and guide future efforts in understanding the role of the DMN for the brain (Box 1), for consciousness, and its proper methodological topographic–dynamic investigation (Box 2).

Declaration of Competing Interest

None.

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- • of special interest
- •• of outstanding interest.

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In this article, the authors propose that anticorrelations involving the default mode network (DMN) represent a fundamental physiological signature—or "prior"—of conscious brain function, which highlights the role of this network in actively shaping brain modularity. This framework supports the idea that the presence of DMN anticorrelations may be necessary for sustaining subjective experience.

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